

# Institutional Dynamism in Entrepreneurial Ecosystems

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**Abstract** Entrepreneurs are involved in entrepreneurial ecosystems. These ecosystems incorporate institutions that interact with entrepreneurs in different ways depending on the stage of the business life cycle the entrepreneur is in. By incorporating institutional theory and business life cycle theory to the analysis of ecosystems, we propose which kind of institution will answer the entrepreneurial question addressing who is there to help their venture depending on the stage of the life cycle it is in. This paper seeks to provide guidance on the institutional support available to increase a venture's chance of success and improve the efficiency of the resource base of ecosystems.

**Keywords** Entrepreneurial ecosystems · Institutions · Business life cycle  
High growth firms

**JEL Codes** L26 · M21 · M38

## 1 Introduction

Entrepreneurs—or those who seek to generate value through the creation or expansion of economic activity and through identifying and exploiting new products, processes, or markets (Ahmad and Seymour 2008)—devise, create, and develop their projects in a social and geographical context. This context is known in

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the literature as the entrepreneurial ecosystem (Brown and Mason 2017; Spigel 2017; Stam 2015). The literature has revealed that the personality of the entrepreneur and the nature of the activity are not the only reasons behind the success of a venture; there is an ecosystem that helps these ventures to prosper (Autio and Thomas 2013).

In this paper, we argue that there is a series of institutions in these ecosystems that facilitate the relationship between the different actors. These institutions include the law and established procedures that normally generate *tangible* agencies (usually labelled as formal institutions), and also the values, beliefs, and culture of a given community (informal institutions; North 1990). Given the relationship between entrepreneurial ecosystems and institutions, we integrate both of these areas of literature to explore the nature of the kinds of institutions that are involved in an entrepreneurial ecosystem. Our first objective is to propose a taxonomy of the institutions that are present in an entrepreneurial ecosystem. We suggest that there are four main types of institutions: (a) institutions linked to venture creation; (b) support organizations; (c) financing; and (d) infrastructure elements, awards, and grants.

Entrepreneurs look for what the ecosystem can provide them and consequently search for the institutions that can give them and their ventures what they are seeking. However, the nature of these institutions may differ depending on the stage the venture is in. This is why in studying the relationship between entrepreneurial ecosystems and institutions, we go one step further and analyse the relationship from a dynamic perspective. We propose that the role of institutions within an entrepreneurial ecosystem is contingent on what stage of the business life cycle the venture is in.

The life cycle theory of a firm is used to show that successful entrepreneurs grow through different stages.<sup>1</sup> This theory suggests that an entrepreneur's activities, attitudes, and requirements change from stage to stage. Consequently, after defining a taxonomy of institutions that operate in an entrepreneurial ecosystem, we put this taxonomy into a dynamic context to show how, depending on the stage an entrepreneur is in and on the typology of the institution that operates in the ecosystem, the entrepreneur will look for and eventually find different kinds of institutions. The purpose of the paper is not to depict how firms' external knowledge or support requirements change as firms grow, but rather to show how the different types of institutions address the specific needs that firms have in the various stages of the firm's growth cycle.

The contributions of this paper are twofold. First, we help to integrate the literature on entrepreneurial ecosystems and institutions with the aim of contributing to the development of the theoretical framework of the former. The idea is that institutional theory may provide a basis to define and categorize the different actors operating in entrepreneurial ecosystems. Institutions play a determining role

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<sup>1</sup>Broadly speaking, the literature identifies five main stages in the business life cycle: birth, growth, maturity, revival, and decline (Miller and Friesen 1984), although the number of stages is not standardized. Other classifications include only three stages: growth, maturity, and decline (Miles et al. 1993) or other numbers depending on different circumstances (Levie and Lichtenstein 2010).

in the entrepreneurial ecosystem as they are there to help the entrepreneur's venture. Better guidance on available institutional support will increase a venture's chance of success and improve the efficiency of the resource base of ecosystems. Second, and more importantly, we analyse entrepreneurial ecosystems from a dynamic perspective, suggesting that each stage of the business life cycle needs a specific treatment to determine the types of institutions that are the most appropriate. It is important to note that the so-called, 'one size fits all' strategies cannot be applied to firms without taking the context into account. On the contrary, it is necessary to consider the specificities of each stage and act accordingly. The link between institutions and the stage of the firm's lifecycle facilitates the identification of the institution that best fits entrepreneurs' interests and may provide them with the necessary support to launch and develop their business.

## 2 Entrepreneurial Ecosystems

Spilling (1996, 91) introduces the concept of entrepreneurial ecosystem as 'the complexity and diversity of actors, roles, and environmental factors that interact to determine the development of entrepreneurship in a region or locality'. An entrepreneurial ecosystem is a set of individual elements including leadership, culture, capital markets, and open-minded consumers that interact in complex ways (Isenberg 2010) and in a specific geographic area (Cohen 2006), thus influencing the formation and final trajectory of the different actors and potentially the whole economy (Spilling 1996; Iansiti and Levien 2004). There is an emerging body of literature in the field of entrepreneurship dedicated to examining the interaction between entrepreneurs, firms, and the context in which they operate (see Chap. 1), that is, to examining entrepreneurial ecosystems.

Although they have common characteristics (Neck et al. 2004; Isenberg 2010), each entrepreneurial ecosystem is different (Mason and Brown 2014). Based on these differences, we should examine the multiple actions conducted by entrepreneurs and their contribution to the dynamism, performance, and long-term transformation of regional economies (Spilling 1996). Therefore, and for the purpose of this paper, it seems plausible to analyse regions with entrepreneurial activity as a system instead of examining the actions of the actors and organizations involved in the region individually and separately. The latter approach is usually tracked in the entrepreneurship literature (Neck et al. 2004). However, it would be a mistake to believe or think that entrepreneurial ecosystems are the domain of only a few highly economically productive places.

Van de Ven (1993) explains that entrepreneurial ecosystems evolve through a set of interdependent elements that interact to generate the creation of new businesses. The existence of entrepreneurial ecosystems makes sense if they facilitate the development of new businesses and ensure their success. Therefore, governments need to create an ecosystem that supports entrepreneurs and facilitates their growth and the creation of new businesses (Isenberg 2010). Additionally, if we focus on a

special type of company such as start-ups, which traditionally experience difficulties with prospering, entrepreneurship ecosystems facilitate their development, especially in the case of ‘high-growth’ companies (Bahrami and Evans 1995). The literature suggests that entrepreneurial ecosystems provide competitive advantages to partners within the ecosystem (Clarysse et al. 2014).

Entrepreneurial ecosystems are often facilitated by external action, especially by the government (Van de Ven 1993). In this sense, entrepreneurial ecosystems are the result of a ‘smart evolution’ (Isenberg 2011), meaning that the market’s invisible hand joins the government’s deliberate hand. According to Isenberg (2010), government authorities can design entrepreneurial ecosystems that later become self-sufficient. At this point, the presence of government agencies would be significantly reduced, although they would not disappear completely. Policy makers seek to stimulate knowledge ecosystems in hotspots of technology with the aim of generating value networks that help participating companies gain a competitive advantage (Clarysse et al. 2014).<sup>2</sup>

Ács et al. (2014) redefine the concept of entrepreneurial ecosystems, creating the new concept of National Systems of Entrepreneurship (NSEs) and providing an approach to characterize them. An NSE is the ‘dynamic, institutionally embedded interaction by individuals between entrepreneurial attitudes, abilities, and aspirations, which drives the allocation of resources through the creation and operation of new ventures’ (17).

The basis for these NSEs is the availability of resources to provide opportunities at the individual level with the purpose of creating new companies, regulated by the specific institutional characteristics of each country. In these NSEs, institutions generate and regulate the activity.

### 3 Elements Within an Entrepreneurial Ecosystem

Having verified the existence of environmental elements that facilitate the work of the entrepreneur, we can now go on to analyse what they are. Reviewing several studies on environmental factors that affect entrepreneurship in different geographical areas, Bruno and Tyebjee (1982) draw up a list of the key factors most commonly cited: the availability of capital, the presence of experienced entrepreneurs, a technically skilled workforce, accessibility to suppliers, access to customers or to new markets, favourable governmental policies, proximity to universities, availability of land or buildings and offices, availability of support services, and attractive living conditions in the area.

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<sup>2</sup>Nonetheless, there are various cases of the lack of effectiveness of public policies. This ineffectiveness is caused by corruption (Estrin et al. 2013), by the way policy is formulated (Arshed et al. 2014), or by the lack of connection between knowledge and business (Clarysse et al. 2014).

Later, in their description of Silicon Valley, Bahrami and Evans (1995) talk about the existence of six elements in the entrepreneurial ecosystem: universities and research institutes, a private investment market, support infrastructure, a base of talent, an entrepreneurial spirit, and a leading group of users. Universities and research institutes not only provide the scientific and technological training, but also generate links between people that facilitate the creation of new companies. A private investment market whose investment profile differs depending on the phase at which the project is located, delivers funds in different rounds of financing, renders management skills that facilitate the sustainability of projects and provides networking, monitoring of the funding, and even guidelines for the entrepreneur. Support infrastructures are necessary for entrepreneurs to be focussed on their projects and not dissipated in peripheral activities; support include law firms, human resources services, and engineering and design firms to develop prototypes, a base of talent, including both local and foreign talent, is critical to launch and sustain the venture. An entrepreneurial spirit is described as a critical although intangible ingredient of the ecosystem that ends up creating a culture of hard and well-done work, and behaves as a source of inspiration that accepts failure as part of the process. Finally, it is important to have access to a leading group of users—early adopters who are eager to incorporate new technologies and products.

Similarly, in their study of the entrepreneurship ecosystem of Boulder County in Colorado, Neck et al. (2004) also find a number of interrelated elements that facilitate, support, and enhance entrepreneurship. These elements are incubators, spin-offs, formal and informal networks, physical infrastructure, and culture. Incubators are defined as the organizations where entrepreneurs were employed before creating their own company (Cooper 1986). Spin-offs are new companies formed by an individual or group of individuals who leave an existing company and start a new one in the same industry (Garvin 1983) or related sectors [this is the version used by Neck et al. (2004)]. Formal networks include universities, government agencies, professional support services, financial sources, access to qualified personnel, or the presence of large companies. Informal networks comprise, among others, friends, family, and co-workers. Physical infrastructure refers to tangible infrastructure components in the county: communications, access to offices, housing, etc. Finally, Neck et al. (2004) mention the existence of a specific culture: one that makes an organization, an industry, or even a nation, unique (Mintzberg et al. 1998).

Isenberg (2010, 2011) identifies 50 components involved in an ecosystem and groups them into six domains: a culture of entrepreneurship, policies that facilitate entrepreneurship and leadership, access to adequate funding, appropriate human resources, a market receptive to innovative products, and support from infrastructures and institutions. A culture of entrepreneurship creates an attitude that rewards innovation, creativity, and experimentation. A suitable legislation enhances entrepreneurship. Access to funding includes microcredit and private and public investment. Human resources provide skilled or unskilled workers. A market receptive to innovative products facilitates the creation of the appropriate distribution

channels and entrepreneurial networks. Support from infrastructures and institutions (incubators, accelerators, etc.) includes legal, economic, and financial support.

3.1 The Entrepreneurial Ecosystem and Institutions

Institutional theory claims that societies have institutions that facilitate market relations (North 1990). These institutions appear because there is a need to generate market rules and regulations. Although institutions limit actors’ behaviours, they facilitate the progress of society.

Institutional theory is being increasingly used in the academic literature, and its application has proven to be helpful to entrepreneurial research. For instance, Bruton et al. (2010) conducted a thorough review of the previous research that had used institutional theory in entrepreneurship and found up to 80 recent articles published in top business journals.

Several classifications of institutions have been presented. However, the taxonomies of North (1990) and Scott (2007) are the most widely accepted. North (1990) distinguishes between formal and informal institutions while Scott (2007) offers the idea of three supportive pillars: regulative, normative, and cognitive. Peng et al. (2009) integrate both these taxonomies and suggest that they are equivalent (see Table 1).

For simplicity, our approach will follow North’s (1990) terminology. North defines institutions as ‘the rules of the game in a society (...) they structure incentives in human exchange, whether political, social or economic. Institutional change shapes the way societies evolve through time and hence is the key to understanding historical change’ (3), and as mentioned distinguishes between formal and informal institutions. The former refer to established laws and procedures while the latter refer to a specific community’s values, beliefs, and culture. Formal and informal factors constitute the rules of the game in a community and affect the birth and the development of new ventures.

The institutional framework influences the decision to become an entrepreneur. For instance, the existence of entrepreneurial activity in a society is inversely

Table 1 Dimensions of Institutions

| Degree of formality (North 1990) | Examples    | Supportive pillars (Scott 2007) |
|----------------------------------|-------------|---------------------------------|
| Formal institutions              | Law         | Regulative (coercive)           |
|                                  | Regulations |                                 |
|                                  | Rules       |                                 |
| Informal institutions            | Norms       | Normative                       |
|                                  | Cultures    | Cognitive                       |
|                                  | Ethics      |                                 |

Source Peng et al. (2009)

proportionate to the size of the government and is significantly related to the absence of corruption (Aidis et al. 2012). Additionally, the chances of becoming an entrepreneur increase in countries that are more entrepreneurship-oriented and have higher levels of trust and education (De Clercq et al. 2013).

As we said, previous literature supports the existence of a number of similar factors or elements within an entrepreneurial ecosystem. In what follows, we introduce institutional theory into the analysis, studying these environmental factors from the point of view of institutions. More precisely, we analyse whether there are different types of institutions involved depending on which of these factors we are considering. Ecosystems consist of formal and informal institutions (Mason and Brown 2014; Stam 2015).

As we have already mentioned, Bahrami and Evans (1995), Neck et al. (2004), and Isenberg (2010, 2011) have reached similar conclusions; that is, that there are six different kind of institutions involved in the entrepreneurial ecosystem. As can be seen in Table 2, these authors agree on the existence of four groups of factors with similar characteristics within what we have called the formal ecosystem: (a) the existence of institutions related to the process of creating the idea, the generation of the concept, the actual launch of the innovative project, and the existence of a market *prepared* to adopt the product; (b) support agencies (i.e., formal networks dedicated to supporting the creation of new businesses); (c) institutions related to the financing of entrepreneurial projects; and (d) other entrepreneurship support infrastructures not included in the rest of the groups. From this similarity in the nature of the institutions, we suggest classifying formal institutions within the entrepreneurial ecosystem into four groups based on those previously cited. We will use the following labels: (a) institutions linked to venture creation; (b) support organizations; (c) financing; and (d) other infrastructure elements, awards, and grants.

Regarding the informal ecosystem, Bahrami and Evans (1995), Neck et al. (2004), and Isenberg (2010, 2011) identify several factors clearly linked to the informal dimension of institutions, and agree on the existence of two kinds of

**Table 2** Elements within an entrepreneurial ecosystem

| Institutional theory  | Bahrami and Evans (1995)               | Neck et al. (2004)      | Isenberg (2010, 2011)                   |
|-----------------------|----------------------------------------|-------------------------|-----------------------------------------|
| Formal Institutions   | Leader users                           | Incubators              | Innovative products                     |
|                       | Universities and research institutions | Formal networks         | Facilitating policies                   |
|                       | Private financing market               | Spin-offs               | Financing                               |
|                       | Support infrastructure                 | Physical infrastructure | Support infrastructure and institutions |
| Informal Institutions | A talent base                          | Informal networks       | Human resources                         |
|                       | Entrepreneurial spirit                 | Culture                 | Culture                                 |

Source Self-made

informal institutions. The first is related to people and their ties and relationships (determined by human resources and networks), and the second is embedded in the culture of the ecosystem.

In this paper, we concentrate on the formal institutions and leave the informal institutions for future research. The reason for this is that informal institutions are more difficult to systematize, strongly depend on people's attitudes and beliefs, and would require an empirical approach that is beyond the scope of this paper.

### ***3.2 Start-Ups and High Growth Firms***

At this point, a distinction should be made between the business ventures that interact within the entrepreneurial ecosystems in terms of their speed of development. We can basically differentiate between high growth firms (HGF) and traditional start-ups.

The Organization for Economic Co-operation and Development (Wennberg 2013) defines HGFs as those 'enterprises with average annualised growth greater than 20% per annum, over a three year period (...). Growth can be measured by the number of employees or by turnover'. Therefore, the main characteristic of an HGF is its rapid growth. This growth can be measured in many ways: as relative sales growth over time, in relative sales in the sector or the country, or in absolute terms as growth in the number of employees or sales.

Another distinguishing feature of HGFs is their innovative capability. A difference between more traditional start-ups and HGFs lies in the importance of innovation (usually measured but not limited, via R&D investment and intellectual property registration) for their growth (Coad and Rao 2008; Stam and Wennberg 2009). Despite the existence of numerous ways of measuring innovation, Wennberg's (2013) scientific literature review shows that there is a proven difference between the use of the diverse forms of innovation (process, market, and product) when it comes to distinguishing between an HGF and a traditional start-up of more moderate growth.

### ***3.3 The Business or Organizational Life Cycle***

The development and progression of the firm as a linear sequential process through a number of stages is generally known as the corporate, business, or organizational life cycle model, or organizational growth model (Miller and Friesen 1984, among others).

Numerous authors (e.g., Adizes 1979; Greiner 1972; Scott 1971 in Miller and Friesen 1984; Lyden 1975) have investigated the fact that corporations go through different sequential stages that are progressive, make them change their activities

and attitude, modify their needs, and transform the company and industry structure. The different growth models share three main propositions (Levie and Lichtenstein 2010):

- (1) Just like a growing organism, an organization goes through differentiated stages;
- (2) These stages are sequential, predetermined, and predictable; and
- (3) There is an implicit sequence of progress that leads organizations from a primitive stage to a differentiated, more developed and mature stage (Van de Ven and Poole 1995).

This similarity with the life cycle of organisms has a direct influence on the name given to the process: the business ‘life cycle’ (Kroeger 1974; Lippitt and Schmidt 1967).

Organizational literature has traditionally identified five common stages in this corporate life cycle: birth, growth, maturity, revival, and decline (Miller and Friesen 1984). Quinn and Cameron (1983) synthesize nine previous papers, and generate their own model, concluding that there are four phases in the organizational life cycle: (a) entrepreneurial phase; (b) collectivity phase; (c) formalization and control phase; and (d) elaboration of structure stage. Although the labels used were not particularly appealing, their integration of previous models has a lot of merit. Nevertheless, these models were all theoretical and there were no empirical longitudinal studies to support them (Quinn and Cameron 1983; Miller and Friesen 1984). To overcome this problem, Miller and Friesen (1984) analyse data from 36 firms over 161 periods, assigned characteristics to each of the five stages of the corporate life cycle, and have drawn three interesting conclusions:

- (1) These stages have a certain configuration, meaning that there is a pattern of different phenomena with properties not derivable from the summation of their parts;
- (2) Each of the phases are in many ways unique; and
- (3) While the stages of the life cycle are internally coherent and very different from one another, they are by no means connected to each other in any deterministic sequence.

Yet, what is involved in these stages? In the first, or birth phase (Scott 1971 in Miller and Friesen 1984), the new firm is ‘attempting to become a viable entity’ (Miller and Friesen 1984, 1162). Here, entrepreneurs face difficult times; firms are young and have simple and informal structures. In the growth stage (Lyden 1975), the company is starting to grow, their distinctive competence has been established, and there is some kind of market success (Miller and Friesen 1984). Structures are larger, and the business aims to grow to an even larger scale. In the maturity phase, there is more formalization and control (Quinn and Cameron 1983), and bureaucratic structures are established. The fourth stage—revival—is a phase of diversification and expansion of the product market scope (Miller and Friesen 1984) and is normally the last stage of the life cycle for many authors. There is a final stage—decline—that happens when markets dry up and firms consequently decline.

Recent literature on firm growth criticises the prominence and widespread acceptance of this life-cycle view. There is a stream of research that currently refutes the linear growth of firms. Garnsey et al. (2006), for example, determine that the study of the growth of new companies suffers from a lack of conceptual models that can filter and assimilate the evidence that enterprises change over time, and conclude that there is no sequence of linear stages in the growth of firms. Additionally, in their investigation of up to 104 articles about the life cycle theory published between 1962 and 2006, Levie and Lichtenstein (2010) conclude that in the models analysed, there was no general agreement as to what a stage is, the number of existing stages, or on how they relate. Furthermore, in their empirical study of a database of Scottish HGFs, Mason and Brown (2013) show that HGFs do not necessarily follow the standard evolutionary path. In this case, growth is episodic and random and does not follow a sequential pattern.

There should then be a distinction in entrepreneurial ventures between small business start-ups, and the more innovative and high-growth entrepreneurship. Small business start-ups would tend to follow a more classical sequential version of the life cycle model, whereas in HGFs the linear and sequential stage-by-stage growth of firms would not be directly applicable.

Despite the critiques mentioned, there is one aspect of the growth model that has been supported by empirical evidence: company growth shows different stages or configurations at different times in its history (Levie and Lichtenstein 2010). In other words, the existence of the life cycle model may be questioned, but the existence of the different stages is empirically proven. This is the basis of our argument: that there are different institutions ready to assist companies in each of these differentiated stages. Consequently, institutions are dedicated to act in the ecosystems to support the dynamism of various stages of new venture creation and growth, regardless of the order of interaction needed by the entrepreneur or the moment in time such interaction is required. Additionally, it is important to note that the existence of these stages does not imply that all companies will necessarily move over time from one stage to the next in a sequential way. This is of special importance for HGFs that, as we have noted, do not necessarily follow a standard evolutionary path.

Having established the stages of the organizational life cycle, we should bear in mind that even before the organization is born, entrepreneurs have to carry out many activities. This is of special importance in the context of entrepreneurial ecosystems because they are not only dedicated to existing firms, but are also involved in the venture creation process. In other words, when discussing entrepreneurial ecosystems, we should remember that ‘the first success of a firm is its birth’ (van Gelderen et al. 2005, 365), and that there are four additional stages, known as ‘pre-start-up phases’, prior to the actual existence of the firm that should be considered. In these four phases, those that are involved in creating businesses are called ‘nascent entrepreneurs’ (Reynolds and White 1992 in van Gelderen et al. 2005). The ‘pre-start-up phases’ should be added to the phases of the firm growth model when considering entrepreneurial ecosystems, as there are institutions in these ecosystems dedicated to giving support to nascent entrepreneurs.

### ***3.4 The Entrepreneurial Ecosystem and the Business Life Cycle***

To our knowledge, there is no attempt in the literature to link entrepreneurial ecosystems with the different stages of the business life cycle (with the exception of Stam and Bosma 2015). Borrowing from Miller and Friesen's (1984) birth, growth, maturity, revival, and decline stages and from van Gelderen et al. (2005), who incorporate previous research to conclude that there are four pre-start-up phases, we propose the following entrepreneurial phases in order to characterize an entrepreneurial ecosystem: (a) intentional; (b) conceptual; (c) seed; (d) start-up; (e) growth; and (f) maturity. That is, we propose to incorporate the four pre-start-up phases (the last coincides with the birth stage of the business life cycle) with the first three stages of the business life cycle. We consider that entrepreneurial ecosystems are not involved, at least in a significant manner, in attending firms in the last two stages of the business life cycle proposed by Miller and Friesen (1984; i.e., revival and decline).

The first or intentional phase concerns the development of an intention to start an enterprise (Krueger et al. 2000). In the second phase, an entrepreneurial opportunity is recognized, and a business concept is developed. In this conceptual stage, the project is therefore basically an idea or concept. Entrepreneurs usually start by finding an opportunity and begin to seek solutions in the form of a product, service, or both, assessing what outcomes to expect. Entrepreneurs in this phase are usually found in research centres and universities, need very little funding, and are basically seeking initial advice on whether their idea has potential (van Gelderen et al. 2005). As time goes by, ideas incorporate a business concept, and projects enter the seed stage. In this third phase, resources are assembled and the organization is created. There are still no sales, and most of the time is spent on research and development. Entrepreneurs continue seeking advice about the most appropriate business model for the company by exploring new alternatives and investigating the potential viability of their venture. Financing needs are growing but are not excessive and are usually covered with insider finance. Funds are dedicated to initial operating expenses. It is the pre-launch stage of a new venture.

In the final phase of the nascent entrepreneur, the organization starts to interact with the market (van Gelderen et al. 2005) and blends into the first phase of the business life cycle. This start-up phase is usually defined as a temporary stage where the company seeks to achieve a scalable business model. The company has already started selling its products or services, although in a restricted manner. Entrepreneurs begin to see the results of market tests about their products. Financing needs for the business grow significantly, especially to obtain the working capital needed to meet sales growth and to cover marketing expenses to increase customer awareness and initial operating expenses. Moreover, market experience forces entrepreneurs to continually adapt their business models to customer requirements. Consequently, entrepreneurs involved in this phase are looking to grow as fast as possible while they improve their offer and obtain financing for their operations.

In the growth stage, the venture is relatively consolidated and seeks to scale-up its activities. While it may still need some financing for working capital, financing is usually allocated to long-term investments that seek the company's consolidation, its expansion, or the restructuring of its operations and growth in local markets.

In the maturity phase, companies seek their final development. This phase usually begins with international growth (in recent times, internationalization often starts even before this phase) and the acquisition of another company to integrate its operations or markets, and culminates with an IPO, or ultimately the sale of the company and replacement of the original shareholders.

### ***3.5 The Entrepreneurial Ecosystem, Institutions, and the Business Life Cycle***

As we have discussed in previous sections, we can find different institutions within the entrepreneurial ecosystem dedicated to facilitating the different needs of an entrepreneur over time. For example, an established firm that is thinking about an IPO would not consider visiting an incubator to develop its business plan. Similarly, a group of young entrepreneurs considering the initiation of their business venture would not visit a bank to issue corporate bonds.

In this section, we build a grid (Fig. 1) that incorporates all the pieces of the puzzle previously described. We argue that depending on the stage an entrepreneur is in and on the typology of the formal institutions that operate in the ecosystem, the entrepreneur will find different institutions. We will consider which institution has a greater influence on the entrepreneur, depending on the stage the entrepreneur is in. We will now describe in detail the type of institutions we can find.

**Institutions Linked to Venture Creation.** These institutions are mainly found in the first stages of the business life cycle (i.e., when the concept is being generated and the business plan is being developed). Among the institutions dedicated to facilitating the creation and launch of new ventures, we can find the following:

- *Nurseries*, usually publicly owned, are aimed at new projects in the very first stages of their life cycle that need space at low cost. The objective is the growth of the project to its next phase.
- *Incubators*, similar to nurseries, facilitate the creation and development of technological entrepreneurial projects prior to their actual launch onto the market and through to early survival (Peters et al. 2004). They are often located in technology centres and have close relationships with local universities.
- *Accelerators* provide rapid growth to projects already launched, shortening the period needed to transform a company in the start-up stage into a profitable and successful company (Kim and Wagman 2014).

| Type of institution                         | Life cycle stage |                                                                                                                                             |                                                  |                                                                                                                                |                |                                             |
|---------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------|---------------------------------------------|
|                                             | Intentional      | Conceptual                                                                                                                                  | Seed                                             | Start-up                                                                                                                       | Growth         | Maturity                                    |
| Institutions linked to venture creation     |                  | Nurseries<br>Incubators<br>Co-working centres<br>Networking areas<br>Networking initiatives                                                 |                                                  | Accelerators                                                                                                                   |                |                                             |
| Support organizations                       |                  | Universities<br>Professional associations<br>Social agents<br>Business schools<br>Research and technology centres<br>Other support agencies | Employment and local development agents          | Clusters<br>Industry and trade chambers<br>Technological parks                                                                 |                |                                             |
| Financing                                   |                  | Insider finance<br>Crowdfunding                                                                                                             | Angel financing<br>Financial institutions        | Venture capital<br>Family office<br>Other investment companies<br>Mutual guarantee societies<br>Financial support institutions | Private equity | Alternative stock markets<br>Stock exchange |
| Infrastructure elements, awards, and grants |                  | Awards                                                                                                                                      | International institutions<br>Other institutions | Support lines<br>Internationalisation aids                                                                                     |                |                                             |

Fig. 1 Institutions and the Business Life Cycle (Source Self-made)

- *Co-working centres* are places where entrepreneurs in the seed and start-up stages share certain common areas, generating synergies between projects in diverse sectors.
- *Networking areas* are places to generate networking to improve business models and companies' commercial activities. Finally,
- *Networking initiatives* are like start-up weekends in which entrepreneurs from different fields and with different abilities and skills work collaboratively to create a start-up, and where networking among entrepreneurs, coaches, mentors, and specialists in different fields facilitates innovation and the development of high-impact ventures.

**Support Organizations.** These institutions facilitate the entrepreneurial process, giving support to entrepreneurs (Ferrary and Granovetter 2009; Bresnahan and Gambardella 2004). They include the following:

- *Universities:* Public and private universities normally have various kinds of advisory services and training in entrepreneurship for projects in the conceptual and seed stages (Swamidass 2013; Levie 2014).
- *Employment and local development agents,* who are workers of local corporations or related to local authorities, and whose main mission is to collaborate in the promotion and implementation of active employment policies related to venture creation.
- *Clusters* are a group of companies, institutions, and universities that share an interest in a particular economic and strategic sector located in the same geographical area. Therefore, clusters within the ecosystem help companies that are entering the growth stage.
- *Professional associations,* which are organizations of employers, are usually grouped by industry and carry out actions aimed at certain groups of entrepreneurs, helping them from the seed to the growth stage.
- *Social agents* (employer associations and unions) are organizations representing the social and collective interests of economic agents. They provide advisory services and training programmes for entrepreneurs and unemployed people normally in the creation stages.
- *Business schools:* Academic institutions that cover the training needs of managers and senior professionals and provide specialized courses aimed at entrepreneurs and business creation.
- *Industry and trade chambers:* Public corporations whose mission is to represent, promote, and defend the general interests of trade and industry and provide services supporting entrepreneurs in the seed, start-up, and growth stages.
- *Research and technology centres* are usually non-profit entities created for the purpose of contributing to the overall benefit of society and improving the competitiveness of enterprises through the generation of technological knowledge in the early stages of the business cycle.
- *Technological parks* (or science and technology parks) are physical spaces and facilities that stimulate and manage knowledge and technology flows between universities and research institutions, companies, and markets. They promote the

creation and growth of innovative companies, generating knowledge, collaborative networks, and technology transfer.

- *Other support agencies*: Those not included in the above categories that also participate in the entrepreneurship ecosystem. Some examples include public institutions dedicated to helping young entrepreneurs, those specialized in specific industries, or municipality- owned agencies dedicated to entrepreneurs.

**Financing Institutions.** As mentioned above, the type of financing that can be achieved by an entrepreneur is also different depending on the stage the venture is in (Berger and Udell 1998).<sup>3</sup> We can find the following:

- *Insider finance*, also known as ‘Friends, Family, and Fools’: These investors support entrepreneurs in their first steps. In these early days, investors have to be found among relatives, friends, or some ‘fools’ who trust entrepreneurs.
- *Angel finance* or *Business angels*: These are investors in the capital stock of the company. The term ‘angels’ refers to the fact that they contribute in the early stages, especially in the start-up phase, with something more than capital (with contacts, experience, management, etc.) and facilitate the expansion of the company to the next phase (Shane and Cable 2002; Mason 2011; Mason and Harrison 1997).
- *Venture capital* are funds of companies that invest in the initial stages of a start-up and participate from the early stages to the growth phase (Carter and van Auken 1994).
- *Private Equity*: These are private equity funds that invest from the growth to the maturity stage. There are four types of private equity funds: (1) growth funds that invest in relatively mature companies that seek financing to restructure their operations or to enter new markets; (2) buyout funds, or funds specialized in acquisition financing (thus, they often need additional leverage); (3) rescue funds specialized in the redirection of a company that is going through problems; and (4) substitution funds that substitute an existing investor in a company (a private investor, a fund, etc.).
- *Alternative stock markets* are dedicated to small cap companies looking to expand. These normally have less liquidity than the stock markets and provide funding, visibility, liquidity, and valuation to SMEs.

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<sup>3</sup>There is a general consensus that depending on the stage of the life cycle they are in, small firms present different capital structures and sources of financing. The most widely accepted view (Gregory et al. 2005) of small-business capital structure is that of Berger and Udell’s (1998) financial growth cycle model. This model explicitly suggests that the optimal capital structure evolves with changes in firm size, age, and the information available. According to this model, very small firms (among which we find start-ups) are financed with insider and angel finance; small firms with a limited track record are financed by venture capital companies; firms of medium size with some track record are also financed with venture capital and medium-term financial institutions; and large firms with a known track record are financed with public equity and long-term financial institutions.

- *Stock exchanges* are regulated markets where securities are traded. Subject to regulation (such as the Securities Market Act), they are supervised by specialized agencies (like the SEC).
- *Family offices*: These are companies that manage part of the equity of private families. They provide investment management and other services (e.g., tax or legal services) to families. When the company manages the assets of more than one family, it is known as a multi-family office. These family offices participate in companies and are present from the seed to the growth stage.
- *Financial institutions*: These are companies that offer financial services to businesses and consumers. Two types of financial institutions can be distinguished: those that give credit (banks, financial institutions, etc.), and those that manage risks (insurance company).<sup>4</sup> Due to the uncertainty and risk of the early stages, these institutions prefer to finance ventures from the start-up stage onwards.
- *Other investment companies* are entities that focus on investing funds, especially in the growth stage, rather than on lending.
- *Mutual guarantee societies*: These societies are business entities whose primary function is to grant guarantees and other collateral to facilitate SME operations management (i.e., to help access to bank financing, import and export guarantees, guarantees in tenders and auctions, etc.).
- *Financial support institutions*: These institutions are governmentally-owned companies that aim to participate in private companies to finance their operations. These institutions have different funds and investment programs depending on the stage companies are in and can participate as investors or by granting loans (usually in the form of equity loans).
- *Crowdfunding*: This is a financing system consisting of many investors pooling funds for a project (Vachelard et al. 2016; Vismara 2016). In crowdfunding, a large number of investors contribute with small amounts of money to facilitate the achievement of a certain objective of the entrepreneur, normally in the seed stage. There are several types of crowdfunding platforms: (1) capital or equity crowdfunding, where investors receive a percentage of the company's stock for their money; (2) reward crowdfunding, where a CD, book, etc., is received in exchange for the money provided, depending on the type of project being financed; (3) pre-purchase crowdfunding, where investors finance the actual production of goods and receive these products in exchange for their investment once the production is completed; (4) donation crowdfunding, where the money is donated for nothing; and (5) loan crowdfunding, where investors lend money to the entrepreneurs.

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<sup>4</sup>In this paper, we will only consider financial institutions that provide specific services to entrepreneurs.

**Infrastructure Elements, Awards, and Grants.** Within these institutions we can find the following (Hsu 2006):

- *Awards* are a term used for institutions that give awards to projects related with entrepreneurship, usually in the early stages.
- *Support lines* are requested and processed by the banks. These are special financing lines that fund investment projects in the early stage.
- *International institutions aids* include EU aids to entrepreneurs in all stages, for example.
- *Internationalization aids* are institutions that give grants and support to promote the internationalization of enterprises.
- *Other institutions* that are not included in the previous categories, like local government aids to increase competitiveness, or subsidies to develop knowledge and technology in a certain area.

Figure 1 shows the organizations representing formal institutions. We should make an additional distinction between the organization and the programs or support services they provide. The reason is that different organizations may provide similar services. For instance, universities that are situated in the conceptual model distinctly to accelerators, incubators, or business schools can offer any or all of the support programs and services these other institutions provide. Similarly, nurseries and incubators offer open spaces to brainstorm, access to networks of business partners, one-on-one or group coaching, access to executive education programs, or support for funding through seed funds, business angels' networks, and so on, just like universities do. Likewise, banks can offer services from early establishment through IPO depending on the banks' market orientation. In conclusion, although formal institutions are very different as organizations, some of the services provided can be more similar.

## 4 Conclusions

This paper has analysed the relationship between entrepreneurial ecosystems and institutions in order to propose a taxonomy that identifies which institutions are more suitable at the different stages of the venture life cycle the firm is in. In recent years, we have witnessed the explosion of a myriad of public and private institutions aimed at stimulating entrepreneurship all around the world. It is true that these institutions may constitute valuable assistance for individuals or groups that decide to start a new venture. However, it is not easy for them to identify the institution that best fits their interests and may provide them with the support necessary to launch their venture. As a consequence, any guidance in this matter will be helpful and will facilitate an adequate assignment of resources.

Our research has identified the different agents that provide support to entrepreneurs within entrepreneurial ecosystems, taking into account that the level of

development of the venture usually determines the value that these agents may provide. With this aim in mind, we depart from the concept of company life cycle. The firm usually evolves through different stages, and we match these stages with the institutions that participate in the entrepreneurial ecosystem. Even if company growth did not follow a linear sequential process, stages would exist, and firms in these stages would still share common characteristics that make them suitable to receive similar aid from formal institutions.

Our taxonomy is valuable both for the individuals that have created (or are in the process of creating) a new venture and for policymakers. From the entrepreneurs' point of view, an adequate knowledge of the whole set of public or private agents that constitute the ecosystem may provide a clear picture of the possibilities available for receiving different types of help (technical or administrative support, financial resources, or even new shareholders that assist in the development and growth of the firm).

From a public policy perspective, the information that we provide in this research is helpful for two reasons. First, a detailed knowledge of the entrepreneurial ecosystem may facilitate policymakers' action, paying special attention and providing resources to these stages less covered by private or other public institutions. Second, it is widely known that public resources are scarce. Thus, it is necessary to decide where to allocate these resources. It is also known that entrepreneurship is not always equally desirable (Baumol 1990), and governments may be more interested in promoting certain kinds of ventures (e.g., quality or high-impact entrepreneurship that is considered to generate higher levels of employment or economic growth).

This paper opens several promising avenues for further research. First, it portrays a theoretical view of how organizations representing formal institutions interact with entrepreneurs within the entrepreneurial ecosystem, and how institutions are dedicated to helping entrepreneurs depending on the stage the venture is in. However, it is not grounded in empirical evidence. Future research should incorporate the empirical component to test how accurate our theoretical predictions are. For example, we may find that only a small proportion of high-tech firms use incubators or accelerators even if they are in the start-up stage, or that firms not only use or interact with universities in their conceptual/seed stages—which is true for university spin-offs.

Second, the life cycle view has been questioned in recent literature. Nonetheless, the existence of the stages as moments in the growth of companies where firms share common characteristics and needs is empirically proven. Therefore, more research is needed on the evolution of firm growth from this non-sequential perspective.

Finally, informal institutions are not included in the framework. Their inclusion could represent the icing on the cake for the link between institutions, ecosystems, and the business stage. In this research, a more sociological approach could be useful.

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## References

- Ács ZJ, Autio E, Szerb L (2014) National systems of entrepreneurship: measurement issues and policy implications. *Res Policy* 43(3):476–494
- Aldizes I (1979) Organizational passages—diagnosing and treating lifecycle problems of organizations. *Org Dyn* 8(1):3–25
- Ahmad N, Seymour R (2008) Defining entrepreneurial activity: definitions supporting frameworks for data collection. OECD Statistics Working Paper
- Aidis R, Estrin S, Mickiewicz TM (2012) Size matters: entrepreneurial entry and government. *Small Bus Econ* 39(1):119–139
- Arshed N, Carter S, Mason C (2014) The ineffectiveness of entrepreneurship policy: is policy formulation to blame? *Small Bus Econ* 43(3):639–659
- Autio E, Thomas L (2013) Innovation ecosystems: implications for innovation management. In: Dodgson M, Phillips N, Gann DM (eds) *The Oxford handbook of innovation management*. Oxford University Press, Oxford
- Bahrami H, Evans S (1995) Flexible re-cycling and high-technology entrepreneurship. *Calif Manag Rev* 37:3
- Baumol WJ (1990) Entrepreneurship: productive, unproductive, and destructive. *J Polit Econ* 98(5):893–921
- Berger AN, Udell GF (1998) The economics of small business finance: the roles of private equity and debt markets in the financial growth cycle. *J Bank Finance* 22(6):613–673
- Bresnahan T, Gambardella A (eds) (2004) *Building high-tech clusters: Silicon Valley and beyond*. Cambridge University Press
- Brown R, Mason C (2017) Looking inside the spiky bits: a critical review and conceptualisation of entrepreneurial ecosystems. *Small Bus Econ*. doi:<https://doi.org/10.1007/s1187-017-9869-3>
- Bruno AV, Tyebjee TT (1982) The environment for entrepreneurship. In: *Encyclopedia of entrepreneurship*. Prentice-Hall, Inc., Englewood Cliffs, pp 288–315
- Bruton GD, Ahlstrom D, Li HL (2010) Institutional theory and entrepreneurship: where are we now and where do we need to move in the future? *Entrepreneurship Theory Pract* 34(3):421–440
- Carter RB, van Auken HE (1994) Venture capital firms' preferences for projects in particular stages of development. *J Small Bus Manage* 32(1):60
- Clarysse B, Wright M, Bruneel J, Mahajan A (2014) Creating value in ecosystems: crossing the chasm between knowledge and business ecosystems. *Res Policy* 43(7):1164–1176
- Coad A, Rao R (2008) Innovation and firm growth in high-tech sectors: a quantile regression approach. *Res Policy* 37(4):633–648
- Cohen B (2006) Sustainable valley entrepreneurial ecosystems. *Bus Strategy Environ* 15(1):1–14
- Cooper AC (1986) The role of incubator organizations in the founding of growth-oriented firms. *J Bus Ventur* 1(1):75–86
- De Clercq D, Lim DS, Oh CH (2013) Individual-level resources and new business activity: the contingent role of institutional context. *Entrepreneurship Theory Pract* 37(2):303–330
- Estrin S, Korosteleva J, Mickiewicz T (2013) Which institutions encourage entrepreneurial growth aspirations? *J Bus Ventur* 28(4):564–580
- Ferrary M, Granovetter M (2009) The role of venture capital firms in Silicon Valley's complex innovation network. *Econ Soc* 38(2):326–359
- Garnsey E, Stam E, Heffernan P (2006) New firm growth: exploring processes and paths. *Ind Innov* 13(1):1–20

- Garvin DA (1983) Spin-offs and the new firm formation process. *Calif Manag Rev* 25(2):3–20
- Gregory BT, Rutherford MW, Oswald S, Gardiner L (2005) An empirical investigation of the growth cycle theory of small firm financing. *J Small Bus Manage* 43(4):382–392
- Greiner L (1972) Evolution and revolution as organizations grow. *Harvard Bus Rev* 50(4):37–46 (July–August)
- Hsu DH (2006) Venture capitalists and cooperative start-up commercialization strategy. *Manage Sci* 52(2):204–219
- Iansiti M, Levien R (2004) Strategy as ecology. *Harvard Bus Rev* 82(3):68–81
- Isenberg D (2010) How to start an entrepreneurial revolution. *Harvard Bus Rev* 88(6):40–50
- Isenberg D (2011) The entrepreneurship ecosystem strategy as a new paradigm for economy policy: principles for cultivating entrepreneurship. Babson Entrepreneurship Ecosystem Project. Babson College, Babson Park
- Kim JH, Wagman L (2014) Portfolio size and information disclosure: an analysis of startup accelerators. *J Corp Finance* 29:520–534
- Kroeger CV (1974) Managerial development in the small firm. *Calif Manag Rev* 17(1):41–47
- Krueger NF, Reilly MD, Carsrud AL (2000) Competing models of entrepreneurial intentions. *J Bus Ventur* 15(5):411–432
- Levie J (2014) The university is the classroom: teaching and learning technology commercialization at a technological university. *J Technol Transfer* 39(5):793–808
- Levie J, Lichtenstein BB (2010) A terminal assessment of stages theory: introducing a dynamic states approach to entrepreneurship. *Entrepreneurship Theory Pract* 34(2):317–350
- Lippitt GL, Schmidt WH (1967) Crises in a developing organization. *Harvard Bus Rev* 47:102–112
- Lyden FJ (1975) Using Parsons' functional analysis in the study of public organizations. *Adm Sci Q* 20(1):59–70
- Mason C (2011) Business angels. In: *World encyclopedia of entrepreneurship*. Elgar original reference. Edward Elgar, Cheltenham
- Mason C, Brown R (2013) Creating good public policy to support high-growth firms. *Small Bus Econ* 40(2):211–225
- Mason C, Brown R (2014) Entrepreneurial ecosystems and growth oriented entrepreneurship. Paper prepared for a workshop of the OECD LEED Programme and the Dutch Ministry of Economic Affairs, The Hague, Netherlands, 7 November 2013
- Mason CM, Harrison RT (1997) Business angel networks and the development of the informal venture capital market in the UK: is there still a role for the public sector. *Small Bus Econ* 9(2):111–123
- Miles G, Snow CC, Sharfman MP (1993) Industry variety and performance. *Strateg Manag J* 14(3):163–177
- Miller D, Friesen PH (1984) A longitudinal study of the corporate life cycle. *Manage Sci* 30(10):1161–1183
- Mintzberg H, Ahlstrand BW, Lampel J (1998) *Strategy safari: the complete guide through the wilds of strategic management*. Financial Times, Harlow
- Neck HM, Meyer GD, Cohen B, Corbett AC (2004) An entrepreneurial system view of new venture creation. *J Small Bus Manage* 42(2):190–208
- North DC (1990) *Institutions, institutional change and economic performance*. Cambridge University Press
- Peng MW, Sun SL, Pinkham B, Chen H (2009) The institution-based view as a third leg for a strategy tripod. *Acad Manage Perspect* 23(3):63–81
- Peters L, Rice M, Sundararajan M (2004) The role of incubators in the entrepreneurial process. *J Technol Transfer* 29(1):83–91
- Quinn RE, Cameron K (1983) Organizational life cycles and shifting criteria of effectiveness: some preliminary evidence. *Manage Sci* 29(1):33–51
- Scott WR (2007) *Institutions and organizations: ideas and interests*. Sage Publications, Thousand Oaks
- Shane S, Cable D (2002) Network ties, reputation, and the financing of New Ventures. *Manage Sci* 48(3):364–381

- Spigel B (2017) The relational organization of entrepreneurial ecosystems. *Entrepreneurship Theory Pract* 41(1):49–72
- Spilling OR (1996) The entrepreneurial system: on entrepreneurship in the context of a mega-event. *J Bus Res* 36(1):91–103
- Stam E (2015) Entrepreneurial ecosystems and regional policy: a sympathetic critique. *Eur Plann Stud* 23(9):1759–1769
- Stam E, Bosma N (2015) Local policies for high-growth firms. In: *Oxford handbook of local competitiveness*. Oxford University Press, pp 286–305
- Stam E, Wennberg K (2009) The roles of R&D in new firm growth. *Small Bus Econ* 33:77–89
- Swamidass PM (2013) University startups as a commercialization alternative: lessons from three contrasting case studies. *J Technol Transfer* 38(6):788–808
- Vachelard J, Gambarra-Soares T, Augustini G, Riul P, Maracaja-Coutinho V (2016) A guide to scientific crowdfunding. *PLoS Biol* 14(2):e1002373
- Van de Ven H (1993) The development of an infrastructure for entrepreneurship. *J Bus Ventur* 8(3):211–230
- Van de Ven AH, Poole MS (1995) Explaining development and change in organizations. *Acad Manag Rev* 20(3):510–540
- van Gelderen M, Thurik R, Bosma N (2005) Success and risk factors in the pre-startup phase. *Small Bus Econ* 24(4):365–380
- Vismara S (2016) Equity retention and social network theory in equity crowdfunding. *Small Bus Econ* 46(4):579–590
- Wennberg K (2013) Managing High-Growth Firms: a literature review. International Workshop on Management and Leadership Skills in High-Growth Firms, Warsaw, 6 May 2013. OECD

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